PROGRAMMING FUNDAMENTALS I

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ALGORITHM

An algorithm can be defined as “a process that performs some sequence of operations in order to solve a given problem”

- The word Algorithm is said to have come from the title Al-Khwārizmī. Muḥammad ibn Mūsā al-Khwārizmī a Muslim mathematician, astronomer (850 C.E)
- Systematic procedures that computers require.

GOOD ALGORITHM & BAD ALGORITHM

- Cutting a piece of plywood with a pair of scissors

example of bad algorithm.

- How many times does 8 go in the number 100

a)

100/8

STEPS IN AN ALGORITHM

- The 4 primary steps in algorithms are
  - Input
  - Output
  - Process
  - Storage

Before solving an algorithm we must identify these steps , write the input (Given) and output (Results)

ASSIGNMENTS COMMANDS

Syntax X = 5Y + 16

On the left side of =, we put the name of a variable and on the right side we put a value or an expression.

Each variable refers to a unique location in computer memory that contains a value.

Interpretation

An assignment is executed in two steps :

1-evaluation of the expression found on the right side.

2-setting the returned value in the memory cell corresponding to variable.

Example Let SideSize=15 Let Area=SideSize * SideSize

ASSIGNMENTS COMMANDS

VARIABLES

ASSIGNMENT IN COMPUTER SCIENCE AND EQUALITY IN MATHEMATICS

TRACING AN ALGORITHM

The purpose of tracing an algorithm is to ensure that it works

- This is a paper test. it should be completed before writing the computer code

- Tracing involves

- Executing the sequence of instructions with a sample set of values

- Computing the value of each variable after each instruction is executed

- Checking for the correct result

TRACING AN ALGORITHM

Step 1 - Number every instruction in the algorithm

Step 2 – Make a table

The first column of the table indicates which instruction has been executed

Subsequent columns list all the variables of the algorithm (Givens, Results)

TRACING AN ALGORITHM

Step 3 – Complete the table

Each column of the table represents a variable

Start at the first line of your algorithm. Identify what will happen to each variable as that instruction is executed

Change any values which the instruction changes and leave all other columns blank

ALGORITHM FOR AVERAGE OF THREE NUMBERS

Trace Average Algorithm using the numbers 24, 31, and 35

APPROACHES IN ALGORITHM DESIGN

Algorithms can be classified into types depending on the approach they use to solve a problem.

- Helps gain insight about an algorithm and understand its general approach.
- Helps get solution for problems that can be solved using the general approach.
- Some algorithms cannot be classified , whereas some can be classified as a combination of approaches

RANDOMIZED ALGORITHMS

- Choose a random number and work with that.
- Rely on statistical properties of random numbers.
- Example QuickSort
- Assume we are given a list to sort,

  8 2 5 6 10 4

- We randomly choose number 5 as a pivot

RANDOMIZED ALGORITHMS

- Move all elements less than 5 to the left , and greater than 5 to the right

- Sort left and right lists using same method. i.e Choose a random pivot and sort left and right

4 2 5 6 10 8 2 4 5 6 10

8 2 4 5 6 8 10

RECURSION

- When an algorithm step , calls / invokes the execution of the algorithm again with different input.

It is said to be recursive

DIVIDE AND CONQUER

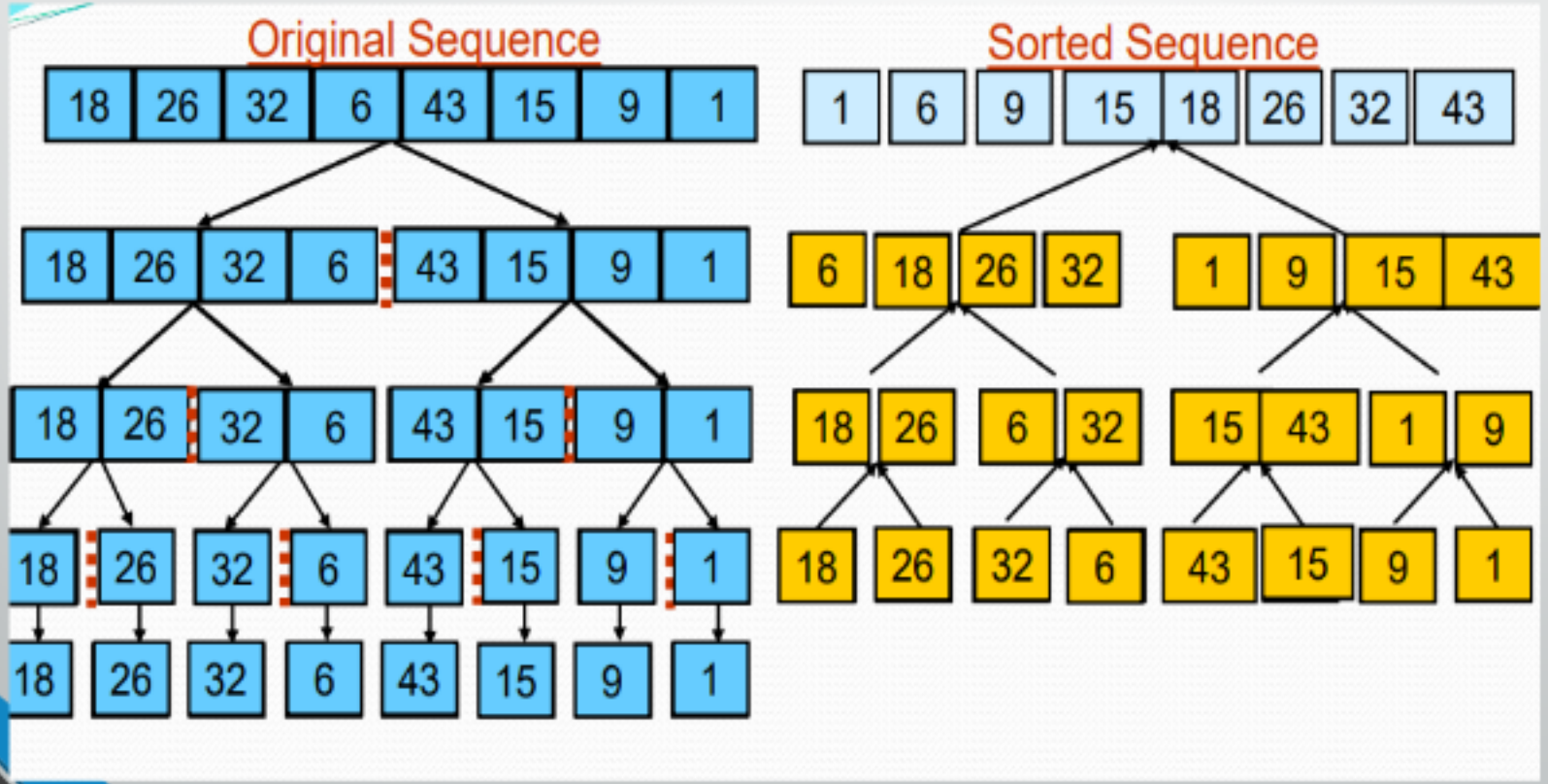
3 steps

Divide - break the problem down

Conquer -Solve the smaller problem

Combine - Merge smaller solutions to solve the bigger problem

MERGE SORT


Original Sequence
Sorted Sequence
18 26 32 6 43 15 9 1
1 6 9 15 18 26 32 43
18 26 32 6 43 15 9 1
6 18 26 32 1 9 15 43
18 26 32 6 43 15 9 1
18 26 6 32 15 43 1 9
18 26 32 6 43 15 9 1
18 26 32 6 43 15 9 1
18 26 32 6 43 15 9 1

DYNAMIC PROGRAMMING SOLUTIONS

Similar to Divide and conquer – both break problems into sub problems

In divide and conquer we use recursion to solve sub problems they are independent of each other. In Dynamic Programming , sub problems are dependent on each other.

Better than divide and conquer since it does not solve

EXAMPLE OF FIBONACCI SERIES

The Fibonacci Sequence is the series of numbers:

0, 1, 1, 2, 3, 5, 8, 13, 21, 34, ...

The next number is found by adding up the two numbers before it.

The 2 is found by adding the two numbers before it (1+1)

Similarly, the 3 is found by adding the two numbers be

EXAMPLE OF FIBONACCI SERIES

Thus to calculate the Nth term of the Fibonacci sequence we use the formula

$F(n) = F(n-1) + F(n-2)$ with $[F(1) = 0 , F(2) = 1]$

GREEDY ALGORITHMS

Greedy algorithms make decisions that look best at the moment.

They make decisions that are locally optimal in the hope that they will lead to globally optimal solutions.

Not exhaustive and does not give accurate answers to many problems

But when it works it is the fastest method

ANALYSIS OF ALGORITHM

While designing or applying an algorithm we must

understand how the algorithm will perform

Algorithm's performance mainly based on how fast

it is

how much space / resources does it use.

ANALYSIS OF ALGORITHM

Definition: Analysis of algorithms is the theoretical study of computer program performance and resource usage, and is often practised abstractly without the use of specific programming language or implementation

Predict the performance of different algorithms in order to guide program design.

Choose between different available options (example sorting)

ANALYSIS OF ALGORITHM

Inefficient algorithms can significantly impact system performance.

In time-sensitive applications an algorithm taking too long to run can render its result outdated or useless.

Over consuming resources.

Performance of algorithms differs based on the inputs

WEST CASE ANALYSIS

passed.

Generally we have 3 categories

Best

Average

Worst

Performance varies significantly between them, we are most

interested in the worst case